

RetainFlow Automation: Customer Churn Prediction

Data Science Bootcamp | Capstone Project | Sohee Lee



What is Customer Churn?



Customer Cancellation

When customers stop using a service



Business Impact

Directly affects growth and profitability



Critical Management

Predicting churn is essential for success

Why Churn Matters

→ Business Impact

Fierce competition → churn = direct **revenue loss**

New customer = **5x more** expensive than retention (HBR)

5% churn reduction = +25-**95% profit**(Bain&Co)

Churn Prediction + Automation

Real-time retention triggers when customers show churn signals



The Critical Gap

Prediction $\neq$ Action

No link to real retention strategy

Manual Response = Slow Death

Marketing & support react too late

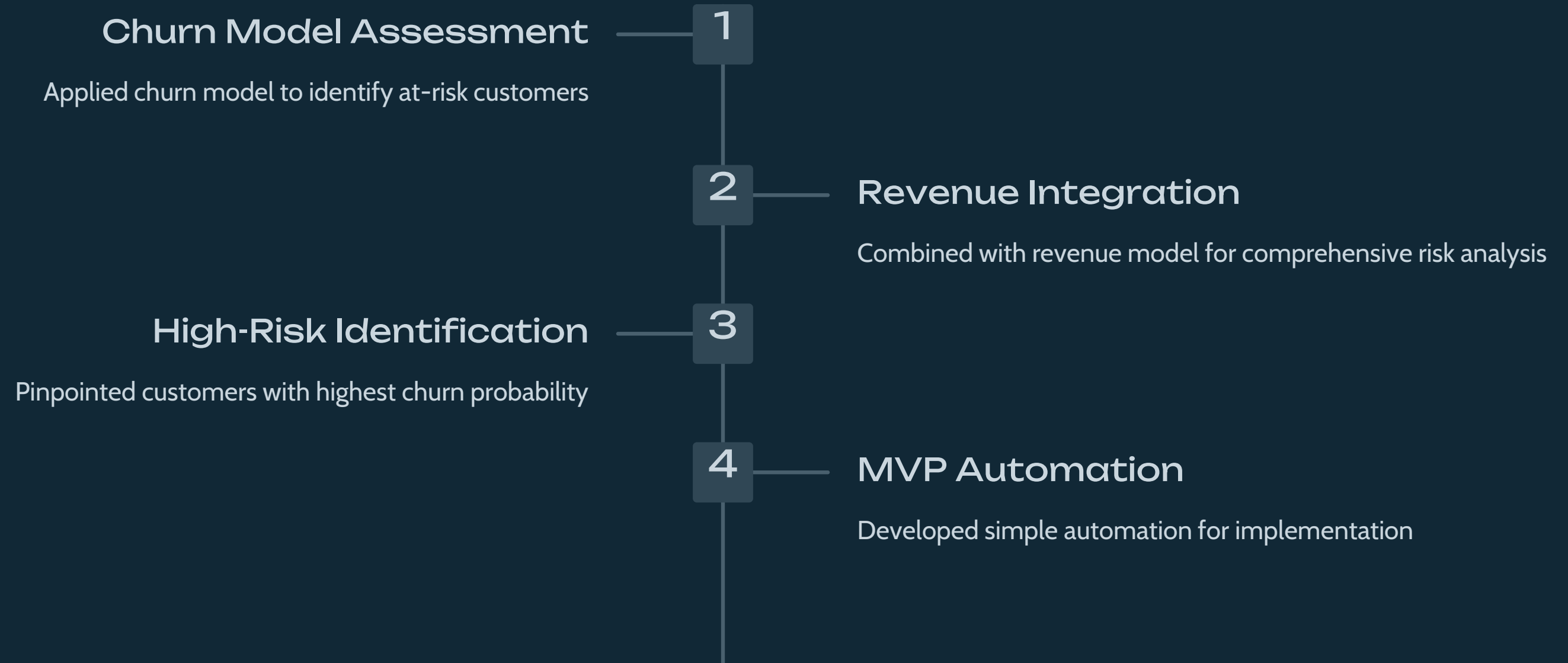
Golden Time Lost

Customers slip away during delays

Prediction is not enough. Action must be automated.



Churn Risk Assessment & Automation



Dataset Overview

7,000+ Customers

Telco industry dataset

Multi-dimensional Features

- Contract terms & payment methods
- Service usage patterns
- Billing history & charges and so on

⚠ **Challenge:** Class imbalance & limited behavioral signals



Machine Learning Framework

Tech Stack

Python 3.10+, scikit-learn, Streamlit deployment

Churn Prediction

RandomForest+Calibration with SMOTE balancing, Kmeans

Revenue Prediction

RandomForest regression with residual learning

Classification Output

Churn probability (0-1)

KMeans segmentation

Accuracy, Recall, Precision, F1, ROC-AUC metrics

Regression Output

Revenue (Total charges) prediction

RMSE & R^2 metrics

Key Data Insights



High-Risk Profile

Short tenure + month-to-month contracts +
e-check payments



Service Gaps Drive Churn

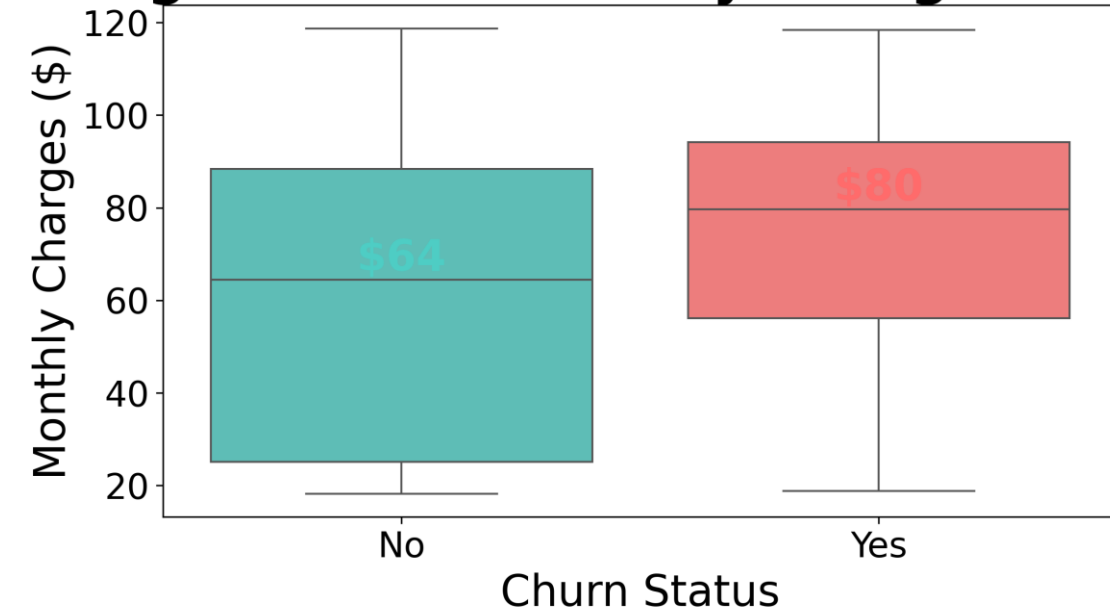
No tech support or online security = higher churn



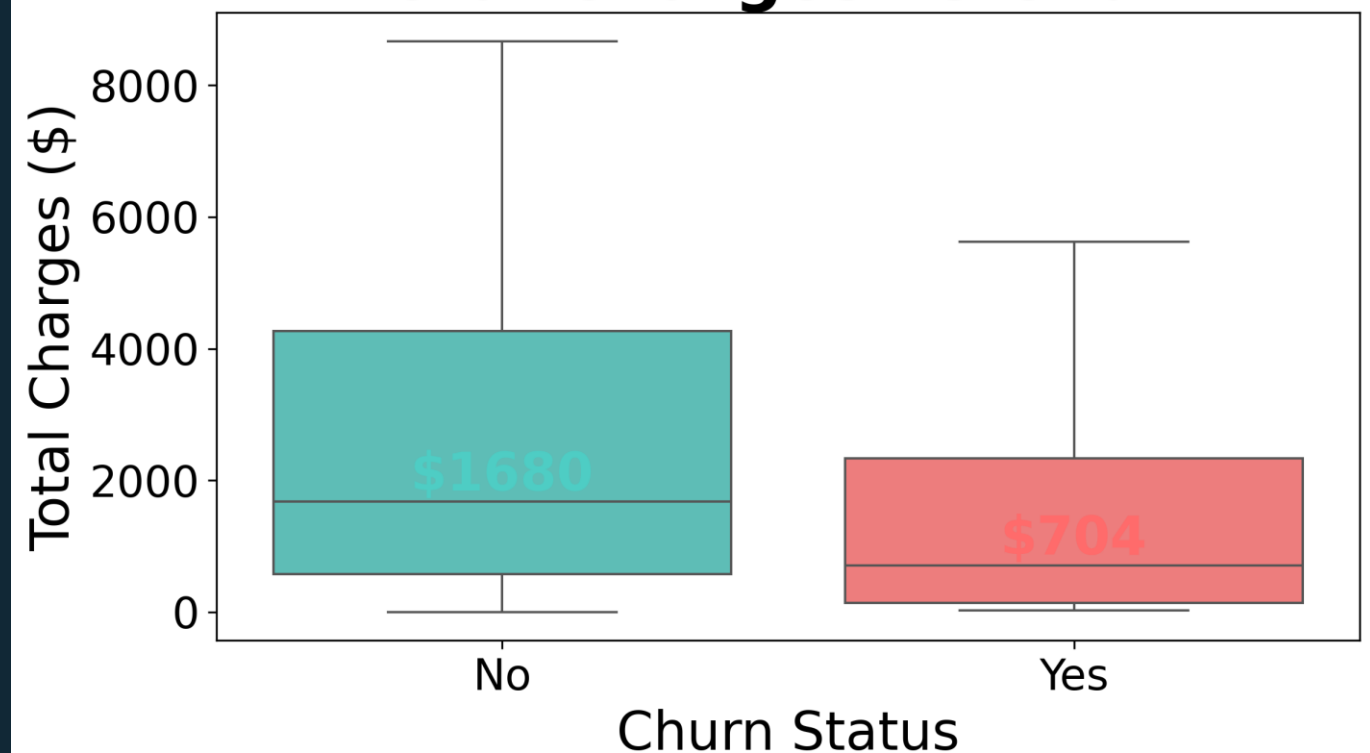
Pricing Paradox

High monthly charges $\uparrow$ churn
High total charges $\downarrow$ churn

Pricing Paradox: Monthly Charges vs Churn



Total Charges vs Churn



Churn Modeling Results

- Key Churn Drivers

Contract Month-to-Month

0.108 importance - Most critical factor

Electronic Check Payment

0.085 importance - Second highest risk

No Online Security

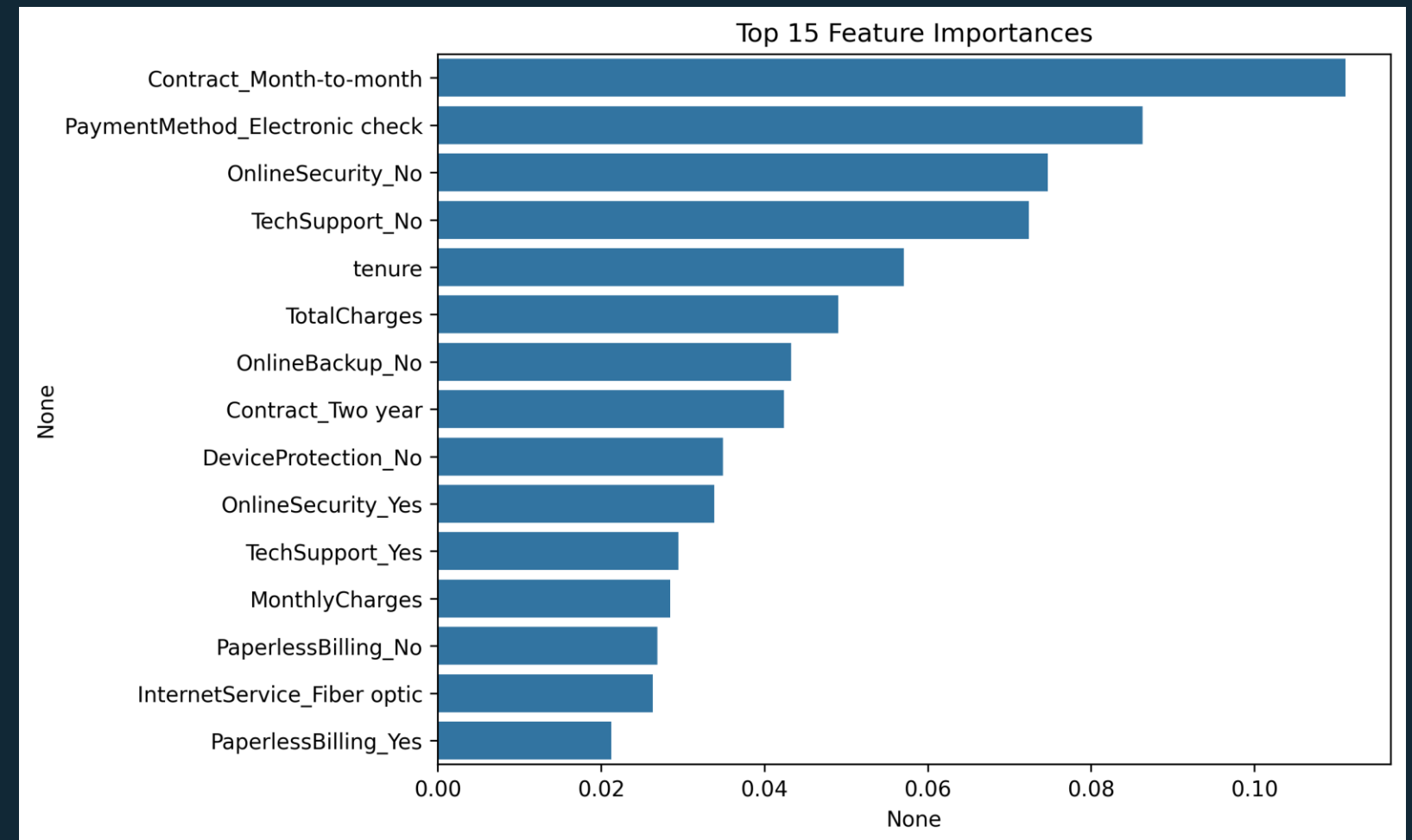
0.072 importance - Third risk factor

No Tech Support

0.068 importance - Fourth risk factor

Short-term Contract + Unstable Payment = Churn Risk

No Security Services = Churn Risk



Churn Modeling Results

- Model Performance Trade-offs

Threshold = 0.5 (Precision = 64%, Recall = 53%)

High Precision

Low Recall

Many churners missed

Threshold = 0.3 (Precision = 55%, Recall = 72%)

Higher Recall

Lower Precision

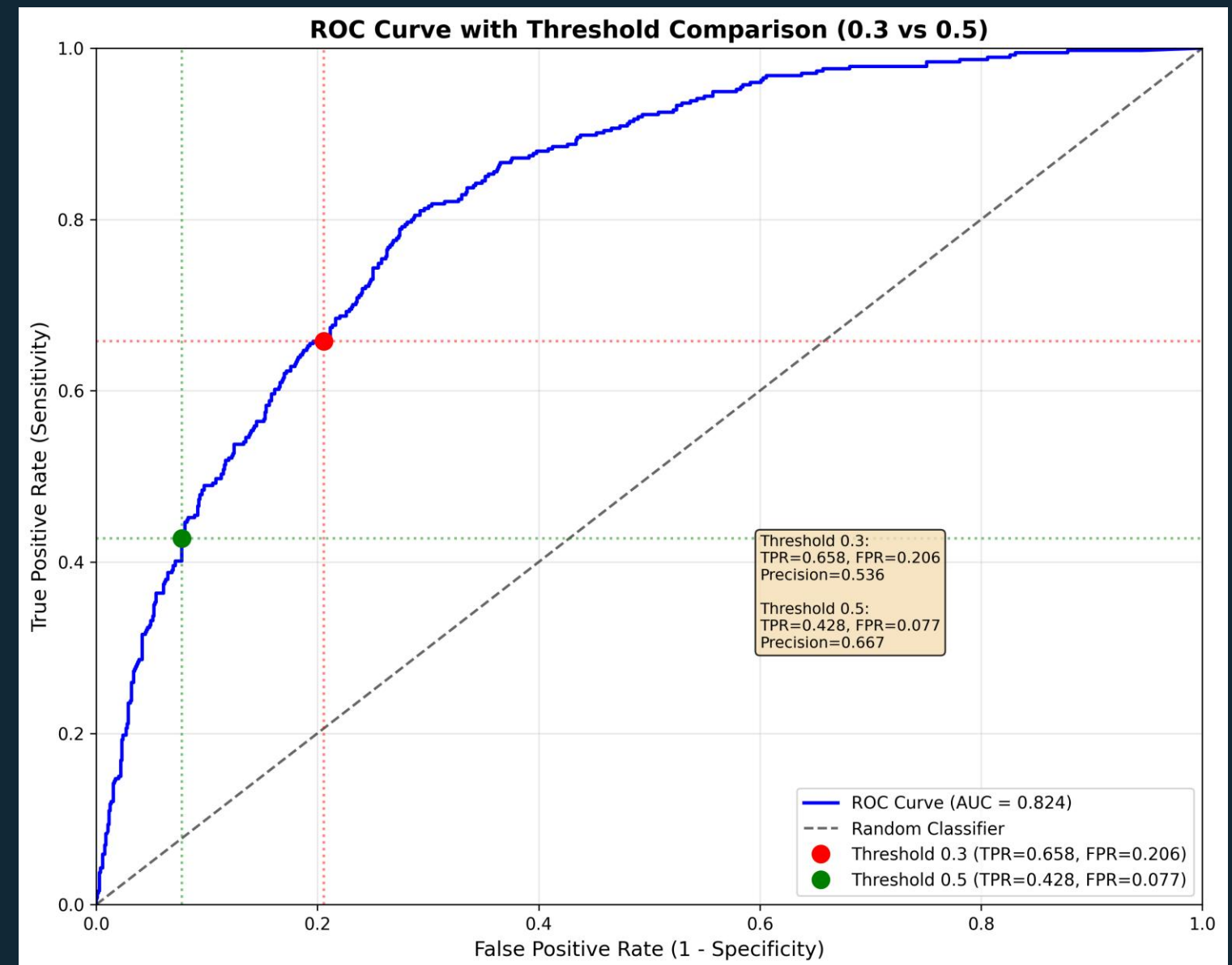
More false alarms

0.81

ROC AUC Score

Strong model performance

Trade-off: Recall ↔ Precision → depends on business goals



Churn Modeling Results

- Customer segmentation

- High Value + Low Risk
VIP Customers - Premium service delivery focus
- Medium Value + Low Risk
Potential Customers - Upselling opportunities
- Low Value + Low Risk
Stable Customers - Maintain basic services
- High Value + High Risk
At-Risk Customers - Immediate retention needed



Monthly charges analyzed against churn probability to enable tailored retention strategies

Revenue Modeling Results

89%

Baseline R^2

Initial linear model performance

55%

Baseline Residual R^2

Initial linear model performance

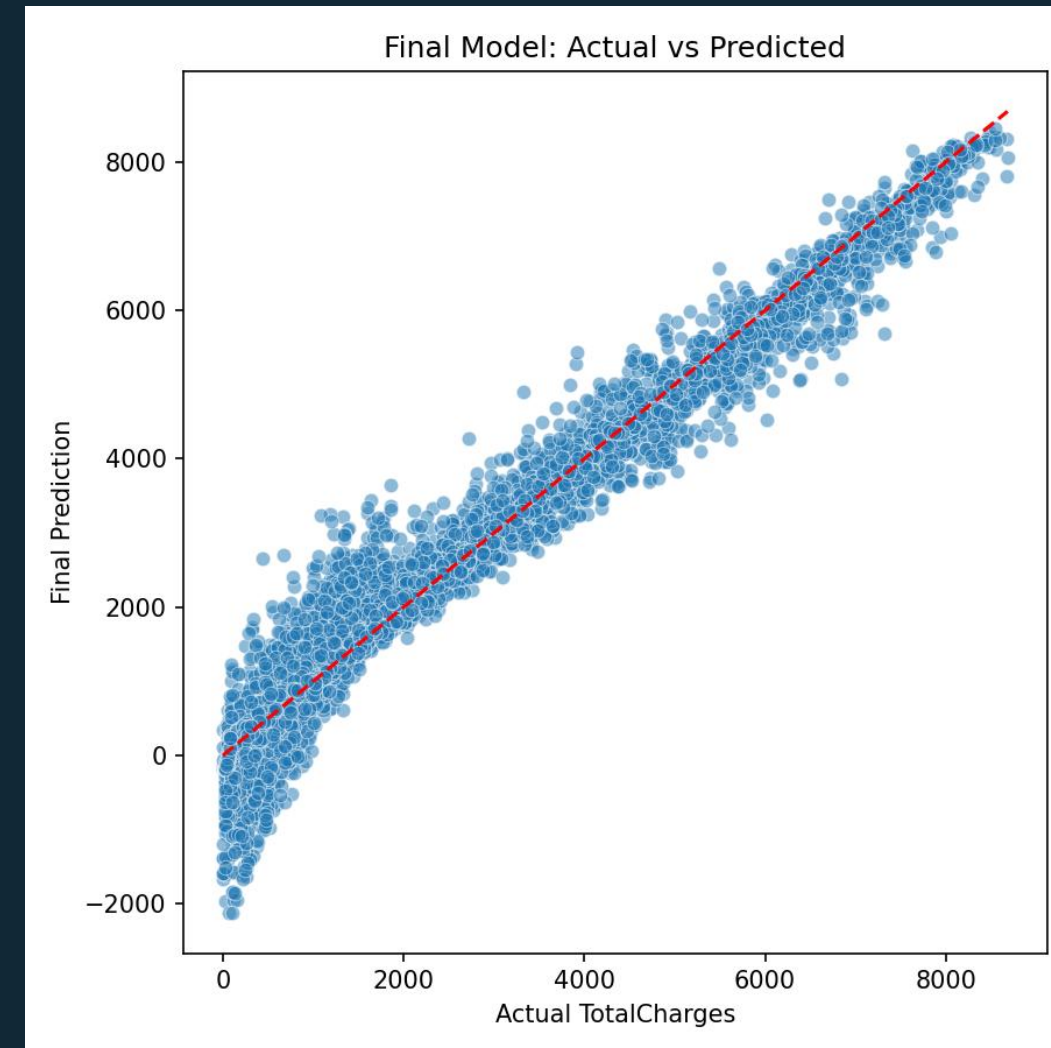
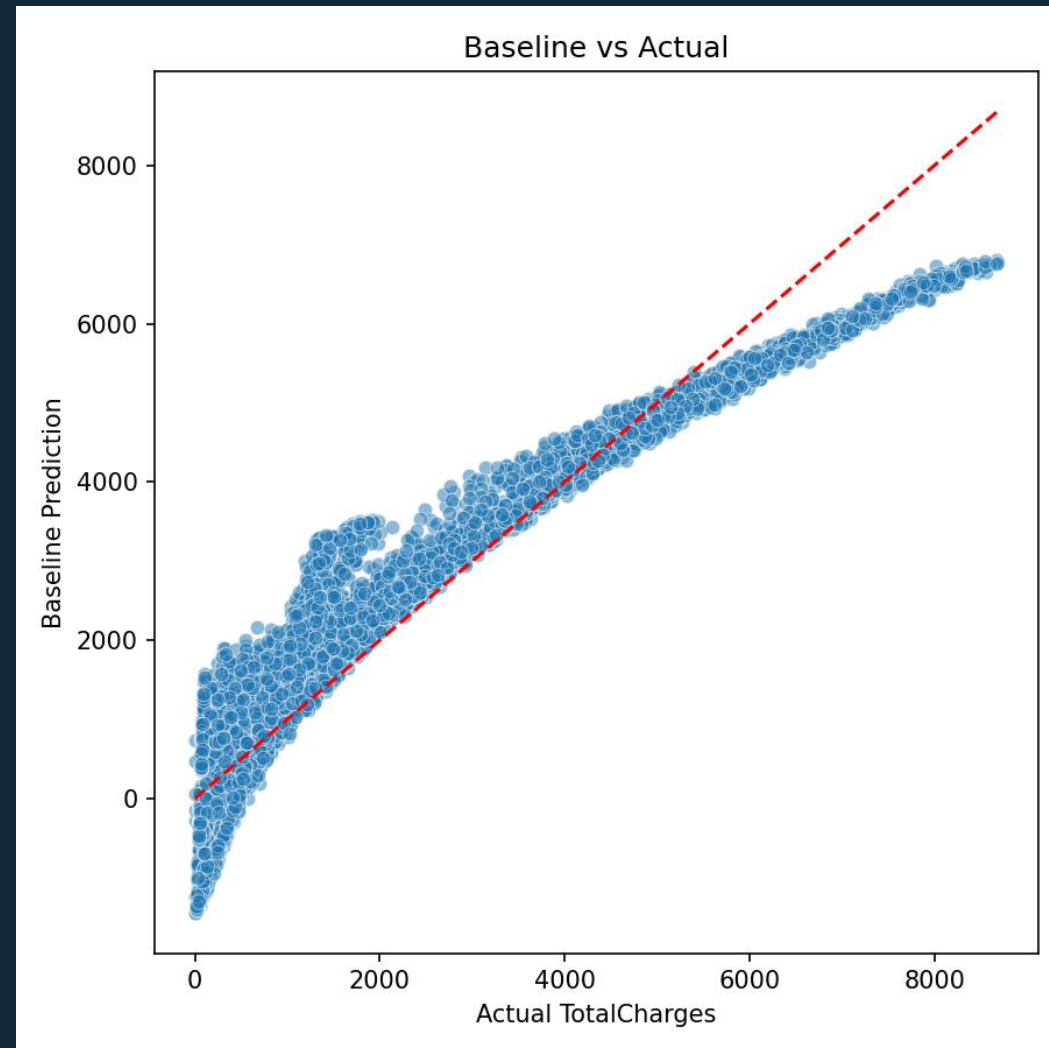
95.5%

Combined R^2

Enhanced predictive accuracy

18%

Final RMSE



Residual learning improved accuracy compared to simple baseline model

Live Demo Integration

Supabase Backend

Real-time data storage

Streamlit Frontend

Interactive predictions

Internal Team Integration

Automated retention actions

Live churn & revenue forecasting → Customer Churn Probability & Segmentation → Instant engagement triggers
→ Top 10 High-Loss Customers (1-Year Projection)

Video Demo

RetainFlow Automation

Problem EDA Modeling/Evaluation Application Outcome

Customer Churn + Revenue Forecasting

✓ Model loaded successfully

✓ Supabase DB connected successfully

Customer CSV upload



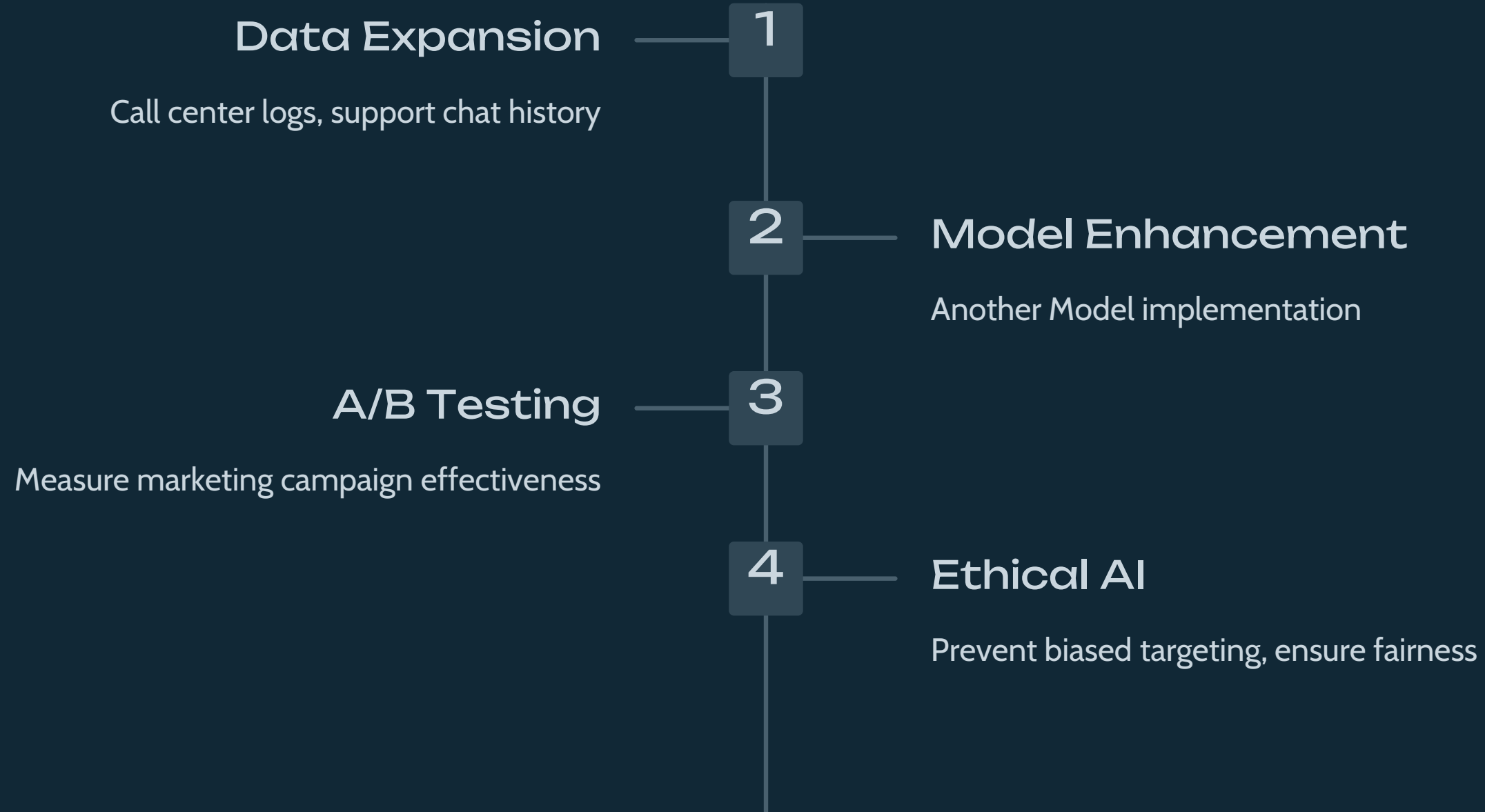
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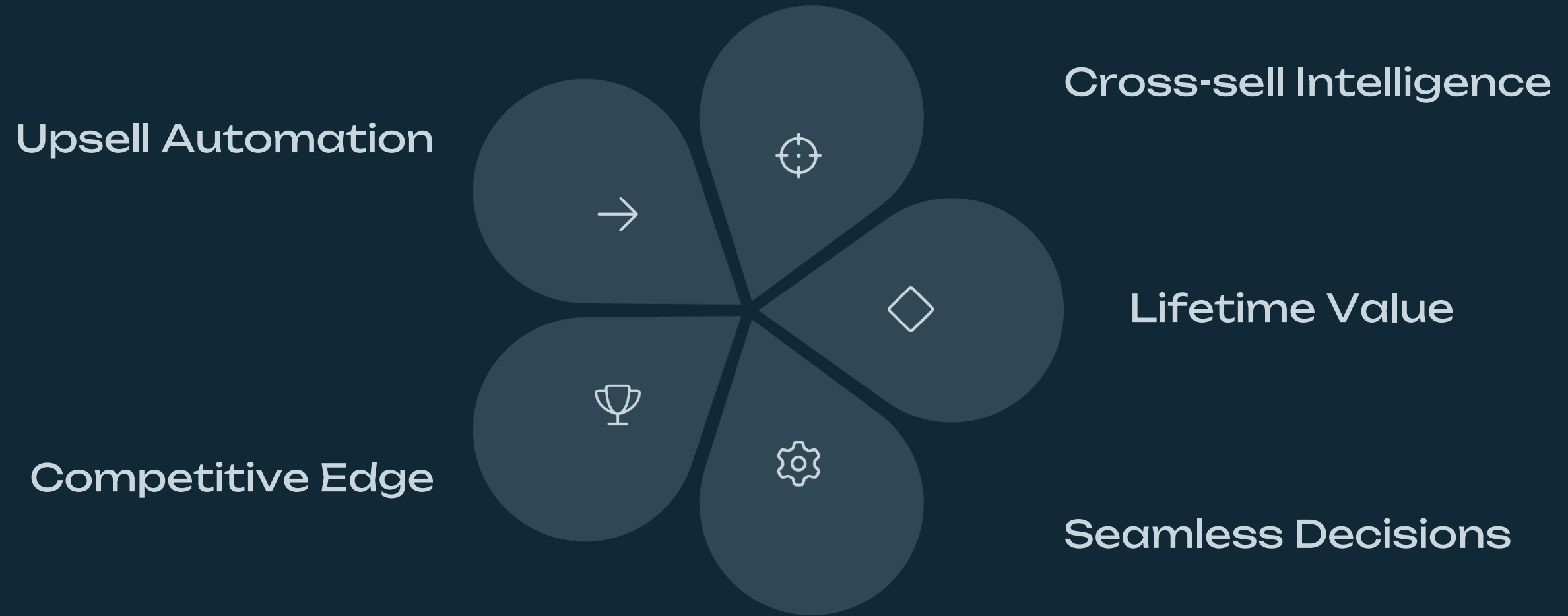
- Churn Probability Prediction
- Customer Segmentation
- Match Personalized Marketing Message

Deploy

Roadmap to Excellence



The Future of Customer Success



One person = productivity of ten with automation

Retention as a competitive advantage

Beyond Prediction

Turn predictions into automated actions.

Thank you

RetainFlow



References

- Johnson, L. M. (2021). *The Customer Retention Handbook: Strategies for Sustainable Growth*. Business Press International. (pp. 45-62)
- Patel, R., & Singh, A. (2019). Predictive Analytics for Customer Churn: A Machine Learning Approach. *Journal of Applied Data Science*, 15(3), 210-225.
- Global Data Insights. (2022). *The Future of Customer Churn: Trends and Forecasts*. Global Data Insights Publications. (pp. 18-24)
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